Full-Mode-Augmentation Tensor-Train Rank Minimization for Hyperspectral Image Inpainting

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***Abstract*— Hyperspectral image (HSI) inpainting is a fundamental task in remote sensing image processing, which is helpful for subsequent applications such as classification and unmixing. Recently, tensor-train decomposition (TTD)-based low-rank methods have achieved great success in image inpainting because of the balanced tensor unfolding and the use of tensor augmentation (TA). However, the TTD algorithm only performs TA on the third mode, and cannot effectively mine the spectral domain information of HSIs. Aiming at this problem, this article extends TA to each mode of the tensor and proposes a full-mode-augmentation TT decomposition (FTTD). More precisely, the HSI is cast along each mode to get a 3-D Nth-order tensor sequence, and all the tensors in the sequence are performed TTD to get $3N$ factor tensors that model the correlation on various modes. Then, the full-mode-augmentation tensor unfolding method is given by performing TT unfolding on each Nth-order tensor. We implement FTTD by minimizing the rank of the unfolding matrix and thus provide the tensor completion framework based on full-mode-augmentation tensor-train rank minimization. Finally, using the framework, we optimize the current two classical iterative algorithms nuclear norm minimization and parallel matrix decomposition. These two algorithms utilize the unfolding matrices in the framework to effectively mine spatial and spectral information, thereby making the restored tensor more approximate to the original HSI. Experiments on various HSIs have shown that the proposed method outperforms compared methods in terms of visual and quantitative measures.**

***Index Terms*— Full-mode-augmentation, hyperspectral image (HSI) inpainting, low-rank tensor completion, tensor augmentation (TA), tensor-train.**

NOMENCLATURE

$\mathcal{H}$	Original and intact HSI.
$\mathcal{T}$	Observed HSI with missing entries.
$\mathcal{X}$	Restored HSI.
$\mathcal{X}(i_1, i_2, \cdots, i_N)$	$(i_1, i_2, \cdots, i_N)$th element of $\mathcal{X}$.
$X_{(k)}$	Mode-k unfolding matrix of $\mathcal{X}$.
$X_{[k]}$	kth TT unfolding matrix of Nth-order tensor $\mathcal{X}$.
$X_{[i,j]}$	jth TT unfolding matrix of the mode-i augmented tensor $\mathcal{X}$.
$\|\mathcal{X}\|_*$	Tensor nulear norm: $\|\mathcal{X}\|_* = \sum_{n=1}^{N} \alpha_n \|X_{(n)}\|_*$.

Manuscript received 16 June 2023; revised 30 August 2023 and 3 November 2023; accepted 1 December 2023. Date of publication 7 December 2023; date of current version 19 December 2023. This work was supported in part by the Natural Science Foundation of Shandong Province under Grant ZR2021MF104 and Grant ZR2021MF113, in part by the National Natural Science Foundation under Grant 62072288, in part by the Key Research and Development Projects of Qingdao Science and Technology Plan under Grant 21-1-2-19-xx, and in part by the Qingdao West Coast New District Science and Technology Plan under Grant 2020-1-6. *(Corresponding authors: Jian-Li Zhao; Mao-Guo Gong.)*

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Digital Object Identifier 10.1109/TGRS.2023.3341030

I. INTRODUCTION

HYPERSPECTRAL images (HSIs) contain rich spatial and spectral information about the target object. It has been applied in many fields such as aerospace, agricultural testing, and medical diagnosis, [1], [2], [3]. However, due to factors such as sensor damage and harsh transmission environments, image information can be lost, and restoring these missing data is considered a key step in subsequent image applications, such as target detection [4], [5], classification [6], [7], and unmixing [8], [9]. In the past two decades, how to complete missing image data has attracted widespread attention from relevant scholars [10], [11], [12].

Currently, HSI inpainting methods can be divided into two categories, namely model driven [13], [14], [15] and data driven [16], [17], [18]. The model driven method has achieved good results in practical applications because it does not require training data and has good interpretability. Tensor, as an effective tool for multidimensional data processing, occupies an important position in model driven image inpainting methods [10], [19], [20]. Utilizing the low-rank characteristics of natural images, the tensor decomposition-based methods transform the image inpainting into a tensor completion problem under linear constraints. In linear algebra, the rank of a matrix is defined as the number of vectors contained in the maximum independent group of its row or column vectors, while there is currently no unified definition of the rank of a tensor. In the past few decades, many researchers have been committed to generalizing the relatively mature matrix theory to tensors for research [21], [22], [23], and they have explored the structure of tensors from various perspectives. Classical tensor decomposition models include CP (CANDECOMP/PARAFAC) decomposition [24], Tucker decomposition [25], and tensor singular value decomposition (T-SVD) [26]. CP decomposition represents a tensor as the sum of a series of rank-one tensors, where the number of rank-one tensors is the rank of the decomposed tensor. Since determining the CP rank of a tensor is an NP-hard problem, the

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rank approximation cannot be obtained progressively. Usually, an iterative method is used to traverse the rank from 1 until a suitable solution is found. The concept of Tucker decomposition was proposed by Tucker in 1966. Tucker decomposes a tensor into a mode product of a core tensor and several factor matrices. Each factor matrix can be viewed as a principal component on the corresponding mode, so Tucker decomposition can be viewed as a higher-order version of principal component analysis (PCA), also known as higher-order PCA. The Tucker mode unfolding of a tensor yields several matrices, and the vector composed of the ranks of these unfolding matrices constitutes the Tucker rank. In fact, when the core tensor of Tucker decomposition is a diagonal tensor, Tucker decomposition degenerates into CP decomposition, so CP decomposition is a special Tucker decomposition. Compared with CP decomposition, Tucker decomposition can use the core tensor to better model the interaction between factor matrices. However, Tucker defines the rank in the form of unfolding matrices, which destroys the structure of the original tensor to a certain extent. To solve this problem, Kilmer et al. [26] proposed a third-order T-SVD which was inspired by the singular value decomposition (SVD) of matrices. T-SVD decomposes a tensor into the product of two orthogonal tensors and a forward diagonal tensor. The tensor rank obtained from T-SVD is called the Tubal rank, which is defined as the number of non-zero diagonal tubes in the tensor. Compared with CP decomposition and Tucker decomposition, T-SVD retains the original structure of the tensor to the greatest extent.

Based on the above three tensor decomposition methods, a large number of image inpainting algorithms have been proposed. Zhao et al. [27] proposed an algorithm for constructing CP decomposition using a hierarchical probability model to solve the problem of determining CP rank. This method adopts complete Bayesian processing by introducing a sparsity-inducing prior on multiple potential factors and introducing appropriate super prior on all super parameters to achieve automatic determination of CP rank. To describe complex hierarchical knowledge with implicit sparse attributes hidden in tensors, Xue et al. [28] coded the structured sparsity of the tensor through multilevel representation, and divided the CP decomposition into two levels of sparsity metrics. The first level is the number of rank-one tensors, and the second level is the local smoothness of the subspace. Since the rank function of a tensor is non-convex and non-smooth, it is difficult to solve it directly. Aiming at this problem, Liu et al. [19] expressed the rank minimization problem of a tensor as the nuclear norm minimization of Tucker mode unfolding matrices. Three image inpainting algorithms based on Tucker decomposition are proposed based on block coordinate descent, smoothing operator, and alternating direction multiplier method, respectively. The algorithm based on nuclear norm minimization involves SVD of matrices, resulting in high time complexity. To solve this problem, Xu et al. [23] proposed a non-convex optimization algorithm for tensor completion. This algorithm decomposes the Tucker mode unfolding matrix into two factor matrices and achieves the low-rank representation of the tensor through the low-rank property of the factor matrices. Zhang and Seron [29] studied low tensor tube rank in video data based on T-SVD and conducted inpainting tasks under random sampling. Experiments have shown that the performance is superior to the algorithms based on Tucker rank minimization. Aiming at the low computational efficiency of T-SVD, Zhou et al. [30] decomposed the third-order tensor into two third-order factor tensors, greatly reducing the computational costs. In order to further improve the effectiveness of image inpainting, many researchers have introduced some other prior knowledge of images based on low-rank tensor completion models. Using the smoothing prior of natural images, Li et al. [31] introduced total variation regularization on the basis of the baseline tensor completion algorithm HaLRTC to enhance the smoothness of images. Zheng et al. [32] exploited the piecewise smoothness prior along all modes of the underlying tensor by introducing smoothness constraints on the factor matrices. Based on the non-local self-similarity prior of natural images, Li et al. [33] introduced a block matching 3-D denoiser (BM3D) using the plug-and-play (PnP) framework on the TMac algorithm. Zhao et al. [34] introduced a convolutional neural network-based denoiser FFDNet using the PnP framework on the basis of the Tubal rank-based baseline tensor completion model, improving the effectiveness of image inpainting by learning a deep prior from a large amount of image data.

In addition to the general image inpainting algorithms mentioned above, a large number of HSI inpainting and denoising algorithms based on low-rank matrix/tensor decomposition have been proposed. These algorithms mostly introduce the unique priors of HSIs based on the common low-rank priors of images to improve their performance. Zhuang and Bioucas-Dias [35] proposed the HSI inpainting algorithm FastHyIn and denoising algorithm FastHyDe. On the one hand, they mine the low-rank structure of HSI by representing the spectral vectors of the image in a pre-learned subspace, and on the other hand, they use groups of similar non-local patches to mine the self-similarity of subspace coefficient images. Considering the global correlation and the non-local self-similarity of remote sensing images, Zheng et al. [36] proposed a novel non-local patch-based tensor decomposition that introduced the fully connected tensor network decomposition to the whole image and its non-local self-similarity groups for image inpainting. Zheng et al. [37] used mode-3 Tucker decomposition to decompose the HSI into a factor tensor representing spatial information and a factor matrix representing spectral information. By adding regularization on two factors, the spatial group sparsity and spectral smoothness of HSIs were improved, respectively. Peng et al. [38] focused on simultaneously developing more accurate approximations to both rank and columnwise sparsity for the low-rank and sparse components. A new method was proposed, which adopts the log-determinant rank approximation and a novel $l_{2,\log}$ norm, to restrict the local low-rank or columnwisely sparse properties for the component matrices, respectively. Although the low-rank representation method mentioned above adds prior knowledge of HSIs from various perspectives, there is currently a lack of designing a tensor decomposition method for HSIs from the perspective of rank optimization.

Recently, tensor completion algorithms based on tensor-train decomposition (TTD) have shown outstanding effectiveness in image inpainting tasks. TT decomposition

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was proposed by Russian scholar Oseledets [39], and Bengua et al. [40] first used this tensor decomposition method for image inpainting in 2017. TT decomposes a tensor into a series of second-order and third-order tensors. Compared with the mode unfolding in the Tucker decomposition, this unfolding method is more balanced, solving the problem that the solved rank is smaller than the actual rank caused by the imbalance between the rows and columns of the unfolding matrices, and thus the restored image contains more detailed information. Considering that TT decomposition is suitable for higher-order tensors, Bengua et al. [40] proposed a tensor augmentation (TA) method called ket augmentation (KA). KA transforms a low-order tensor into a high-order tensor through a structured block addressing method. This operation is of great significance for subsequent image inpainting tasks. This is because the TT unfolding of higher-order tensors will yield more mode matrices, and studying the correlation of each mode matrix can better explore the correlation of the original tensor. Experiments have shown that the tensor completion algorithm based on TT rank minimization greatly improves the effectiveness of image inpainting, and is superior to the tensor completion models based on CP, Tucker, and Tubal rank minimization in most scenarios. Similar to previous image inpainting models, introducing image priors to baseline tensor completion models based on TT rank minimization has been a hot topic in recent years. Ding et al. [41] and Yang et al. [42] enhance the smoothness of images by adding total variational regularization and mixed smoothing regularization, respectively. Zhao et al. [43] integrated global, local, and non-local priors of images by incorporating the BM3D denoiser and the deep learning-based denoiser into the non-convex TT rank minimization tensor completion model. In addition, in view of the importance of order increment in tensor completion models based on TT rank minimization, relevant researchers have improved the TA method. Zhang et al. [11] proposed a multiscale feature tensorization method to help restore images at the feature level. Zhang et al. [44] proposed an overlapping KA method to eliminate the chessboard effect caused by KA.

Unlike color images, each spatial pixel in an HSI contains tens to hundreds of wavebands, and there is a strong correlation between these continuous spectra. Although the above methods have improved the tensor completion model based on TT rank minimization from different perspectives, they all only augment the tensor along the mode-3 direction, that is, only apply TA to the spatial domain. As a result, spectral correlation cannot be effectively mined.

To address this issue, this article focuses on HSI inpainting tasks and proposes a tensor completion framework based on full-mode-augmentation TT rank minimization. The framework first casts the image into higher-order tensors along each mode. Then, for each higher-order tensor, a TT balanced unfolding is performed to obtain several mode matrices. Finally, the completion of tensor missing data is achieved through the rank minimization algorithm of each mode matrix. The specific work is summarized as follows.

1) A full-mode-augmentation TT decomposition (FTTD) is proposed, and its corresponding tensor rank is defined. First, cast the tensor data of HSI along each mode to obtain a 3-D Nth-order tensor sequence. Then, the TT decomposition is performed for each Nth-order tensor in the sequence, resulting in $3N$ factor tensors. Compared with TT decomposition, FTTD can model more correlation of tensor factors.
2) Based on FTTD, a tensor completion framework for HSI inpainting is proposed. First, the full-mode-augmentation TT unfolding of a tensor is defined, and the corresponding relationship between the full-mode-augmentation TT rank and the rank of the unfolding matrix is explained. Furthermore, a tensor completion framework for implementing tensor FTTD by minimizing the rank of the unfolding matrix under linear constraints is proposed. This framework fully exploits the spatial and spectral correlation of HSIs through the correlation of various mode matrices.
3) Two commonly used tensor completion algorithms, nuclear norm minimization and parallel matrix decomposition, are optimized using the tensor completion framework based on full-mode-augmentation TT rank minimization. Under this framework, the two algorithms complete the HSI based on the low-rank properties of $3 \times (N-1)$ full-mode-augmentation TT unfolding matrices. Compared with only using $N-1$ TT unfolding matrices, the algorithms can better capture the correlation of each mode, resulting in a better low-rank approximation for HSIs.

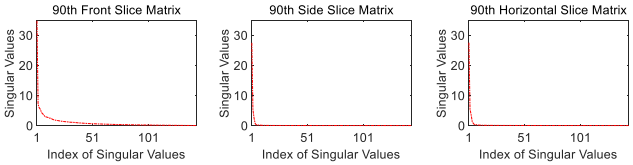

Fig. 1. Singular value distribution of each slice matrix.

The rest of this article is organized as follows. Section II gives some symbols and preliminary knowledge. Section III introduces the proposed tensor FTTD, HSI inpainting framework, and optimization algorithm with a detailed computational complexity analysis. In Section IV, a large number of experiments were conducted to illustrate the advantages of the proposed framework. Section V summarizes the article.

II. Notations and Preliminaries

According to [19], [40], and [45], we have summarized the notations used in this article and their meanings in Nomenclature. Then, several definitions of tensor operations were introduced, including Tucker unfolding, TT unfolding, and TA.

Definition 1 (Tucker Unfolding): Tucker unfolding, also known as mode unfolding, for an Nth-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$, the mode-k unfolding matrix is $X_{(k)} \in \mathbb{R}^{I_k \times (\prod_{i \neq k} I_i)}$, and the element $X_{(k)}(i_k, j)$ in the matrix corresponds to the element $\mathcal{X}(i_1, i_2, \ldots, i_N)$ in the tensor, where $j = 1 + \sum_{s=1, s\neq k}^{N} (i_s - 1) J_s$ and $J_s = \sum_{m=1, m\neq k}^{s-1} I_m$. The N-dimensional vector composed of the ranks of these mode-k unfolding matrices is called the Tucker rank.

Definition 2 (TT Unfolding): TT unfolding, also known as balance unfolding, for an Nth-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$, there are a total of $N-1$ TT unfolding matrices, where the size of the kth unfolding matrix is $(\prod_{s=1}^{k} I_s) \times$

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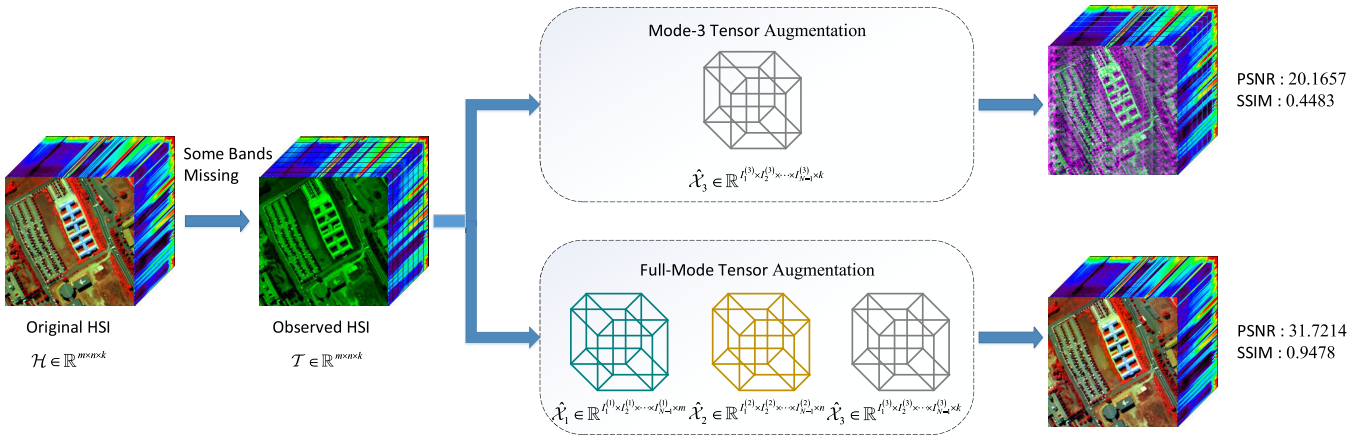

Fig. 2. Comparison between mode-3 augmentation and full-mode-augmentation.

$(\prod_{s=k+1}^{N} I_s)$, and the mapping relationship can be determined by the reshape function in MATLAB, that is, $X_{[k]} = \text{reshape}(\mathcal{X}, \prod_{s=1}^{k} I_s, \prod_{s=k+1}^{N} I_s)$. The N-dimensional vector composed of the ranks of these TT unfolding matrices is called the TT rank.

Definition 3 (TA): Since most multidimensional visual data are of third order in practical applications, direct TT-rank minimization has not yet excavated the estimation potential of TT rank minimization [46]. When handling third-order visual data, such as color images and multispectral images, existing TT-rank-based methods use TA as a tensor order increment preprocessing to transform a lower-order tensor into a higher-order tensor.

The current TA method mainly includes KA [40], and overlapping KA [44], which use different rules to cast a third-order tensor $\mathcal{X} \in \mathbb{R}^{m\times n\times k}$ into an Nth-order tensor $\hat{\mathcal{X}} \in \mathbb{R}^{I_1\times I_2\times\cdots\times I_{N-1}\times k}$ along the mode-3 direction, where $m \times n = I_1 \times I_2 \times \cdots \times I_{N-1}$, $N \gg 3$. According to Definition 2, the number of TT unfolding matrices of tensor $\mathcal{X}$ is 2, while the number of TT unfolding matrices of tensor $\hat{\mathcal{X}}$ obtained using TA is $N - 1$. This can better capture the low-rank structure of the image by exploring the correlation of more unfolding matrices.

III. Proposed HSI Inpainting Framework and Solving Algorithm

A. Problem Formulation and Motivation

Assuming that the third-order tensor data of the observed HSI is $\mathcal{T} \in \mathbb{R}^{m\times n\times k}$, the index set corresponding to the observed entries is Ω, and the restored tensor is $\mathcal{X} \in \mathbb{R}^{m\times n\times k}$, the tensor completion problem of HSI inpainting can be modeled as

$$\begin{aligned} \min_{\mathcal{X}} \quad & \text{rank}(\mathcal{X}) \\ \text{s.t.} \quad & \mathcal{P}_\Omega(\mathcal{X}) = \mathcal{P}_\Omega(\mathcal{T}) \end{aligned} \tag{1}$$

where $\mathcal{P}$ represents the projection operation. The significance of this model is to make the rank of the to-be-completed tensor as small as possible under the condition that the entries in Ω are equal to the corresponding entries in the observed tensor. Due to the non-convex and non-smooth characteristics of the rank function, solving it directly is an NP-hard problem. From this, some approximate solving algorithms have been derived, such as nuclear norm minimization [19] and matrix decomposition [23], [32]. These algorithms implement various tensor decomposition through operations on the unfolding matrix, thereby realizing the completion of missing data.

In order to better restore HSIs, we need to fully explore their low-rank structures. Compared with color images, HSIs provide pixel-level spectral features. To verify the fact that HSIs have both spatial and spectral correlation, we selected an image block of size $144 \times 144 \times 144$ from the remote sensing data Washington DC Mall and then studied the rank of the tensor front slice matrix, side slice matrix, and horizontal slice matrix composed of the pixels in the image block. Since the rank of a matrix is equal to the number of non-zero singular values, we can roughly estimate the rank of the matrix by observing the distribution of singular values. Fig. 1 shows the singular value distribution of the 90th front slice matrix, side slice matrix, and horizontal slice matrix selected, respectively. It can be clearly observed from the figure that all three have low-rank characteristics. Simultaneously mining the correlation between spatial and spectral modes will help improve the effectiveness of image inpainting.

The current tensor completion methods based on TT rank minimization only augment the image along the mode-3 direction. These methods can only mine the spatial correlation of HSIs, which are the low-rank properties of the forward slice matrices, while the spectral correlation of HSIs, which are the low-rank properties of the side and horizontal slice matrices, cannot be effectively mined, thus affecting the effectiveness of image inpainting. Therefore, in order to fully utilize the role of TA and improve the effectiveness of image inpainting, it is necessary to perform full-mode-augmentation on the image.

We illustrate the impact of the TA on the performance of HSI inpainting in Fig. 2. It can be intuitively seen that compared with only augmenting the tensor along the mode-3 direction, full-mode-augmentation exhibits outstanding effects in both visual effects and evaluation metrics.

B. Proposed Tensor Decomposition and HSI Inpainting Framework

For the third-order tensor $\mathcal{H} \in \mathbb{R}^{m\times n\times k}$ used to represent HSI data, the tensor sequence obtained by augmentation along each mode is $\hat{\mathcal{H}} = \{\hat{\mathcal{H}}_1, \hat{\mathcal{H}}_2, \hat{\mathcal{H}}_3\} = \text{FTA}(\mathcal{H})$, where $\hat{\mathcal{H}}_1 \in \mathbb{R}^{I_1^{(1)}\times I_2^{(1)}\times\cdots\times I_{N-1}^{(1)}\times m}$, $\hat{\mathcal{H}}_2 \in \mathbb{R}^{I_1^{(2)}\times I_2^{(2)}\times\cdots\times I_{N-1}^{(2)}\times n}$, $\hat{\mathcal{H}}_3 \in \mathbb{R}^{I_1^{(3)}\times I_2^{(3)}\times\cdots\times I_{N-1}^{(3)}\times k}$ represent the augmentation components

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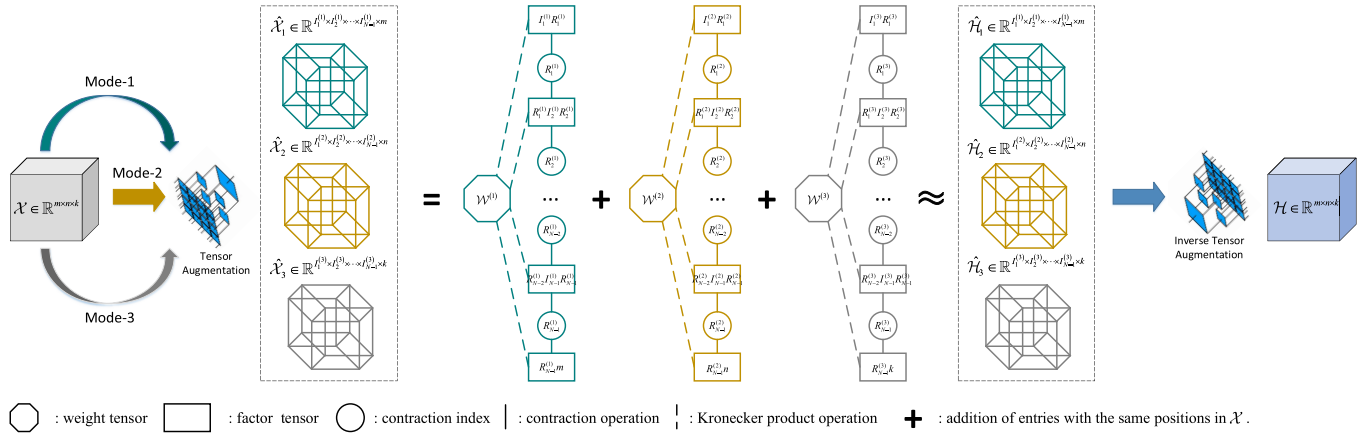

Fig. 3. Full-mode-augmentation TT decomposition.

along mode-1, -2, and -3, respectively. We use tensor $\mathcal{X} \in \mathbb{R}^{m\times n\times k}$ to approximate the original tensor, where the elements in the tensor are represented as

$$\begin{aligned}&\mathcal{H}(i,j,k) \approx \mathcal{X}(i,j,k)\\&= \sum_{p=1}^{3} \Psi_p\left(\mathcal{W}^{(p)}_{1,2,\ldots,N}\mathcal{G}^{(p)}_1\left(l^{(p)}_1\right)\mathcal{G}^{(p)}_2\left(l^{(p)}_2\right),\ldots,\mathcal{G}^{(p)}_N\left(l^{(p)}_N\right)\right)\end{aligned} \tag{2}$$

where $\mathcal{W}$ represents the tensor for assigning weights to each mode; $\mathcal{G}^{(p)}_1(l^{(p)}_1)\mathcal{G}^{(p)}_2(l^{(p)}_2),\ldots,\mathcal{G}^{(p)}_N(l^{(p)}_N)$ is used to fit tensor sequence $\hat{\mathcal{H}}$; For any mode p, Ψ_p represents the mapping relationship from $(l^{(p)}_1, l^{(p)}_2,\ldots,l^{(p)}_N)$ to (i,j,k) in the augmentation mode; $\mathcal{G}^{(p)}_s(l^{(p)}_s), s = 1, N$ is the sth vector of the factor matrix, and $\mathcal{G}^{(p)}_t(l^{(p)}_t), t = 2,\ldots,N-1$ is the tth slice matrix of the third-order factor tensor. For an HSI of size $m \times n \times k$, the FTTD of the corresponding tensor is shown in Fig. 3.

Definition 4 (Full-Mode-Augmentation TT Rank): For a third-order tensor $\mathcal{X} \in \mathbb{R}^{m\times n\times k}$, the rank of its full-mode-augmentation TT rank is defined as a matrix $R \in \mathbb{R}^{3\times(N-1)}$ composed of $3(N-1)$ contraction indices $R^{(i)}_j, i = 1,2,3, j = 1,2,\ldots,N-1$ in order, where $R_{ij} = R^{(i)}_j$ is denoted as $\text{rank}_{\text{FTT}}(\bullet)$.

Definition 5 (Full-Mode-Augmentation TT Unfolding): Let the third-order tensor $\mathcal{X} \in \mathbb{R}^{m\times n\times k}$ have the augmentation components $\hat{\mathcal{X}}_1 \in \mathbb{R}^{I^{(1)}_1\times I^{(1)}_2\times\cdots\times I^{(1)}_N}$, $\hat{\mathcal{X}}_2 \in \mathbb{R}^{I^{(2)}_1\times I^{(2)}_2\times\cdots\times I^{(2)}_N}$, and $\hat{\mathcal{X}}_3 \in \mathbb{R}^{I^{(3)}_1\times I^{(3)}_2\times\cdots\times I^{(3)}_N}$ along mode-1, -2, and -3. The full-mode-augmentation TT unfolding of $\mathcal{X}$ yields $3\times(N-1)$ matrices, where $X_{[i,j]} = \text{reshape}(\hat{\mathcal{X}}_i, \prod_{s=1}^{j} I^{(i)}_s, \prod_{s=j+1}^{N} I^{(i)}_s)$, $i = 1,2,3$, and $j = 1,\ldots,N-1$.

Theorem 1: For any component unfolding matrix $X_{[i,j]}$ of the full-mode-augmentation tensor sequence represented by formula (2), if its rank is r_{ij}, then there exists a TT decomposition of the full-mode-augmentation tensor, such that $R^{(i)}_j \le r_{ij}$.

Based on this theorem, this article proposes a method for achieving FTTD by minimizing the rank of the tensor full-mode-augmentation TT unfolding matrix, and constructs the following tensor completion framework:

$$\begin{aligned}\min \quad & \sum_{i=1}^{3}\sum_{j=1}^{N-1}\alpha_{ij}\text{rank}\left(X_{[i,j]}\right)\\ \text{s.t.} \quad & \mathcal{P}_\Omega(\mathcal{X}) = \mathcal{P}_\Omega(\mathcal{T})\end{aligned} \tag{3}$$

where α_{ij} represents the weight of each unfolding matrix. Sections III-C and III-D will introduce two approximation methods for matrix rank functions to perform the tensor completion task.

C. Nuclear Norm Minimization-Based Optimization Algorithm

Since the rank function of a matrix is non-convex and non-smooth, solving it directly is an NP-hard problem. An effective method is to approximate rank minimization using its convex relaxation nuclear norm minimization [19]. Based on this theory, formula (3) is transformed into the following expression:

$$\begin{aligned}\min_{\mathcal{X}} \quad & \sum_{i=1}^{3}\sum_{j=1}^{N-1}\alpha_{ij}\left\|X_{[i,j]}\right\|_*\\ \text{s.t.} \quad & \mathcal{P}_\Omega(\mathcal{X}) = \mathcal{P}_\Omega(\mathcal{T}).\end{aligned} \tag{4}$$

Introduce auxiliary variable M and construct the following objective function:

$$\min_{\mathcal{X},M_{ij}} \sum_{i=1}^{3}\sum_{j=1}^{N-1}\alpha_{ij}\left\|M_{ij}\right\|_* + \frac{\beta_{ij}}{2}\left\|X_{[i,j]} - M_{ij}\right\|_F^2 + l(\mathcal{X},\mathcal{T}) \tag{5}$$

where $l(\mathcal{X},\mathcal{T})$ is used to express constraint conditions, defined as

$$l(\mathcal{X},\mathcal{T}) := \begin{cases}0, & \mathcal{P}_\Omega(\mathcal{X}) = \mathcal{T}\\ \infty, & \text{otherwise.}\end{cases} \tag{6}$$

1) Update Variable M: Based on the objective function (5), solve the subproblems for each M_{ij} as follows:

$$\min_{\mathcal{X},M_{ij}} \alpha_{ij}\left\|M_{ij}\right\|_* + \frac{\beta_{ij}}{2}\left\|X_{[i,j]} - M_{ij}\right\|_F^2. \tag{7}$$

According to [47], the optimal solution for the closed form of this problem is

$$M_{ij} = \text{D}_{\alpha_{ij}/\beta_{ij}}\left(X_{[i,j]}\right) \tag{8}$$

where $\text{D}_{\alpha_{ij}/\beta_{ij}}(X_{[i,j]})$ represents the operator that performs singular value shrinkage on $X_{[i,j]}$ under threshold α_{ij}/β_{ij}. If the SVD of matrix $X_{[i,j]}$ is $X_{[i,j]} = USV^T$, then

$$M_{ij} = US_{\alpha_{ij}/\beta_{ij}}V^T. \tag{9}$$

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Among them, $S_{\alpha_{ij}/\beta_{ij}} = \text{diag}(\max(\sigma_{ij}^{(k)} - \alpha_{ij}/\beta_{ij}))$ and $\sigma_{ij}^{(k)}$ represent the kth largest singular values of matrix $X_{[i,j]}$.

2) Update Variable $\mathcal{X}$: Consider the following sub-problems when solving $\mathcal{X}$:

$$\min_{\mathcal{X},M_{ij}} \sum_{i=1}^{3}\sum_{j=1}^{N-1} \frac{\beta_{ij}}{2}\left\|X_{[i,j]} - M_{ij}\right\|_F^2 + l(\mathcal{X},\mathcal{T}). \tag{10}$$

For the elements in $\mathcal{X}$ within the Ω set, keep them equal to the corresponding elements in $\mathcal{T}$, and for the elements outside the Ω set, satisfy $X_{[i,j]} = M_{ij}$. Therefore, the solution of variable $\mathcal{X}$ is

$$\mathcal{X} = \mathcal{P}_{\Omega}(\mathcal{T}) + \mathcal{P}_{\Omega^C}\left(\frac{\sum_{i=1}^{3}\sum_{j=1}^{N-1}\beta_{ij}\text{fold}\left(\text{M}_{\text{ij}}\right)}{\sum_{i=1}^{3}\sum_{j=1}^{N-1}\beta_{ij}}\right). \tag{11}$$

Among them, Ω^C represents the complement of set Ω, and fold is the inverse operation of TT unfolding, representing the operation of folding matrix M_{ij} to obtain tensor $\mathcal{M}$.

By summarizing the above solving process, we described the tensor completion algorithm SiLRTC-FTT in Algorithm 1, which achieves full-mode-augmentation TT rank minimization through nuclear norm minimization.

Algorithm 1 Nuclear Norm Minimization-Based Optimization Algorithm for Full-Mode-Augmentation TT Rank Minimization

Input: The observed data $\mathcal{T} \in \mathbb{R}^{m\times n\times k}$, index set Ω.
Parameters: $\alpha_{ij}, \beta_{ij}, i = 1,2,3 j = 1,2,\cdots,N-1$.
Output: $\mathcal{X}$.
Initialization: $\mathcal{X}^0$,with $\mathcal{X}_{\Omega}^0 = \mathcal{T}_{\Omega}$, $l = 0$.

while not converged **do**
  for $i = 1$ to 3 **do**
    for $j = 1$ to $N-1$ **do**
      Unfold the tensor $\mathcal{X}^l$ to $X_{[i,j]}^l$.
      Update M_{ij}^{l+1} from $X_{[i,j]}^l$ by (8).
    end for
  end for
  Update $\mathcal{X}^{l+1}$ from M_{ij}^{l+1} by (11).
end while
return result

D. Parallel Matrix Factorization-Based Optimization Algorithm

Due to the fact that tensor completion convex optimization algorithms involve SVD of matrices, their time complexity is relatively high. Therefore, based on [23], this section expresses the low rank of the full-mode-augmentation TT unfolding matrix through the low rank of the factor matrix. Convert formula (3) into the following objective function:

$$\min_{U_{ij},V_{ij},X_{[i,j]}} \sum_{i=1}^{3}\sum_{j=1}^{N-1} \frac{\alpha_{ij}}{2}\left\|U_{ij}V_{ij} - X_{[i,j]}\right\|_F^2 + l(\mathcal{X},\mathcal{T}). \tag{12}$$

Among them, $X_{[i,j]}$ is the jth TT unfolding matrices of tensor $\mathcal{X}$ augmented along mode-i, U_{ij} and V_{ij} are the two factor matrices of $X_{[i,j]}$, α_{ij} represents the weights of each augmented mode unfolding matrix, and $l(\mathcal{X},\mathcal{T})$ has the same effect as (5).

Algorithm 2 Parallel Matrix Factorization-Based Optimization Algorithm for Full-Mode-Augmentation TT Rank Minimization

Input: The observed data $\mathcal{T} \in \mathbb{R}^{m\times n\times k}$, index set Ω.
Parameters: $\alpha_{ij}, i = 1,2,3 j = 1,2,\cdots,N-1$.
Output: $\mathcal{X}$.
Initialization: $\mathcal{X}^0$,with $\mathcal{X}_{\Omega}^0 = \mathcal{T}_{\Omega}$, $l = 0$.

while not converged **do**
  for $i = 1$ to 3 **do**
    for $j = 1$ to $N-1$ **do**
      Unfold the tensor $\mathcal{X}^l$ to $X_{[i,j]}^l$.
      Update U_{ij}^{l+1} by (14).
      Update V_{ij}^{l+1} by (15).
      Update $X_{[i,j]}^{l+1}$ by (16).
    end for
  end for
  Update $\mathcal{X}^{l+1}$ from $X_{[i,j]}^{l+1}$ by (17).
end while
return result

1) Update Variable U_{ij}: Consider the following sub-problem for solving U_{ij}:

$$\min_{U_{ij}} \frac{\alpha_{ij}}{2}\left\|U_{ij}V_{ij} - X_{[i,j]}\right\|_F^2. \tag{13}$$

Taking V_{ij} and $X_{[i,j]}$ as constants, by taking the derivative of U_{ij} and making the derivative function zero, the following solution is obtained:

$$U_{ij}^{l+1} = X_{[i,j]}^l\left(V_{ij}^l\right)^T\left(V_{ij}^l\left(V_{ij}^l\right)^T\right)^{\dagger}. \tag{14}$$

Among them, "l" represents the number of iterations, and "†" represents the Moore Penrose generalized inverse.

2) Update Variable V_{ij}: The method for solving V_{ij} is similar to U_{ij}, where U_{ij} and $X_{[i,j]}$ are fixed. By taking the derivative of V_{ij} and making the derivative function zero, the following solution is obtained:

$$V_{ij}^{l+1} = \left(\left(U_{ij}^{l+1}\right)^T U_{ij}^{l+1}\right)^{\dagger}\left(U_{ij}^{l+1}\right)^T X_{[i,j]}^l. \tag{15}$$

3) Update Variable $\mathcal{X}$: $\mathcal{X}$ can be obtained by folding and accumulating $X_{[i,j]}$ based on the weight of each unfolding matrix, it can be easily obtained according to (13)

$$X_{[i,j]}^{l+1} = U_{ij}^{l+1}V_{ij}^{l+1}. \tag{16}$$

By combining the weight α_{ij} of each $X_{[i,j]}$ and the constraint condition $l(\mathcal{X},\mathcal{T})$, the following solution is obtained:

$$\mathcal{X}^{l+1} = \mathcal{P}_{\Omega}(\mathcal{T}) + \mathcal{P}_{\Omega^C}\left(\sum_{i=1}^{3}\sum_{j=1}^{N-1}\alpha_{ij}\text{fold}\left(X_{[i,j]}^{l+1}\right)\right). \tag{17}$$

By summarizing the above solving process, we described the tensor completion algorithm TMac-FTT that achieves full-mode-augmentation TT rank minimization through parallel matrix decomposition in Algorithm 2.

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E. Computational Complexity Analysis

We analyzed the computational complexity of SiLRTC-FTT and TMac-FTT on HSI $\mathcal{X} \in \mathbb{R}^{m\times n\times k}$. Let the tensor sequence obtained from the HSI after full-mode-augmentation be $\{\hat{\mathcal{X}}_1, \hat{\mathcal{X}}_2, \hat{\mathcal{X}}_3\}$, where $\hat{\mathcal{X}}_1 \in I_1^{(1)} \times I_2^{(1)} \times \cdots \times I_N^{(1)}$, $\hat{\mathcal{X}}_2 \in I_1^{(2)} \times I_2^{(2)} \times \cdots \times I_N^{(2)}$, and $\hat{\mathcal{X}}_3 \in I_1^{(3)} \times I_2^{(3)} \times \cdots \times I_N^{(3)}$. The time cost of SiLRTC-FTT mainly lies in SVD, and the time complexity of each iteration is $\mathcal{O}(\sum_{i=1}^{3}\sum_{j=1}^{N-1}\min(p_{ij}^2 q_{ij}, p_{ij} q_{ij}^2))$, $p_{ij} = \prod_{k=1}^{j} I_k^{(i)}$, $q_{ij} = \prod_{k=j+1}^{N} I_k^{(i)}$. The time cost of TMac-FTT mainly lies in the decomposition of the unfolding matrix, with a time complexity of $\mathcal{O}(\sum_{i=1}^{3}\sum_{n=1}^{N-1}(L_n^{(i)} R_n^{(i)} S_n^{(i)} + L_n^{(i)}(R_n^{(i)})^2 + (R_n^{(i)})^2 S_n^{(i)}))$ for each iteration, where $L_n^{(i)} = \prod_{k=1}^{n} I_k^{(i)}$, $S_n^{(i)} = \prod_{k=n+1}^{N} I_k^{(i)}$, and $R_n^{(i)}$ are tensor full-mode-augmentation TT ranks.

IV. Numerical Experiments

To verify the effectiveness and performance improvement of the method proposed in this article, experiments were conducted on two real HSIs in this section. The first one is selected from the Pavia University dataset with sizes of 256 × 256 × 81. The other one is selected from the Washington DC Mall dataset, with sizes of 256 × 256 × 144. We used the simplest and flexible TA method, which is the reshape function in MATLAB, to cast the HSI into high-order tensors and verify the effectiveness of the proposed algorithm. All experiments in this article were conducted on an Intel Core i7-10700 microcomputer with 16 GB memory in the MATLAB R2018b environment.

A. Experimental Settings

1) Baseline Approaches: We compared the proposed method with the methods in Table I.

2) Missing Scene: We compared the proposed method with the baseline method in the following three scenarios.

Case 1 (Horizontal Stripes Missing): Randomly select some horizontal slice matrices of HSIs, and the selection rates are set to 0.5, 0.7, and 0.9, respectively. Especially, all elements in the selected matrices will be missing, i.e., set to zero. As a result, all bands in the HSIs show the horizontal stripes missing.

Case 2 (Vertical Stripes Missing): Randomly select some side slice matrices of HSIs, and the selection rates are set to 0.5, 0.7, and 0.9, respectively. Especially, all elements in the selected matrices will be missing, i.e., set to zero. As a result, all bands in the HSIs show the vertical stripes missing.

Case 3 (Spectral Missing): Select some front slice matrices of HSIs and set the elements in the matrices to zero. As a result, several bands of HSIs are missing.

3) Parameter Settings: The selection of parameters has an important effect on the fitting ability of the model [48]. In this article, two parameters, the weight of each mode and the rank of the tensor, need to be considered. For the weight parameters α_{ij} of each mode, set them to $\sum_{j=1}^{N-1}\alpha_{1j} = 0.025$, $\sum_{j=1}^{N-1}\alpha_{3j} = 0.95$, in Cases 1 and 2, and set them to $\sum_{j=1}^{N-1}\alpha_{1j} = 0.25$, $\sum_{j=1}^{N-1}\alpha_{2j} = 0.25$, $\sum_{j=1}^{N-1}\alpha_{3j} = 0.5$ in Case 3. For the full-mode-augmentation TT rank $R_j^{(i)}$ in TMac-FTT, its value is the number of singular values that satisfy the following inequality:

$$\frac{\sigma_k^{[i,j]}}{\sigma_1^{[i,j]}} > \text{th} \tag{18}$$

where $k = 1, \ldots, R_j^{(i)}$, and $\{\sigma_k^{[i,j]}\}$ are assumed to be in descending order. The principle for selecting this parameter is that the lower the rank of the unfolding matrix, the more singular values will be truncated. During the experiment, th was adjusted based on the effectiveness of image inpainting.

This article uses relative change (RelCha) as the stopping criterion for all methods, which is defined as

$$\text{RelCha} = \frac{\left\|\mathcal{X}^{k+1} - \mathcal{X}^{k}\right\|_F}{\left\|\mathcal{X}^{k}\right\|_F}. \tag{19}$$

In all methods, the maximum number of iterations is set to 500, and the iteration is stopped when the RelCha is less than 10^{-4}. The parameters in the compared method are set according to the recommendations of the corresponding paper.

4) Evaluation Metrics: This article uses peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM) to quantitatively evaluate the effectiveness of various methods. Suppose $\mathcal{H} \in \mathbb{R}^{m\times n\times k}$ is the tensor data of the original image and $\mathcal{X} \in \mathbb{R}^{m\times n\times k}$ is the tensor data of the restored image, then PSNR and SSIM are defined as follows:

$$\text{PSNR}(\mathcal{X}, \mathcal{H}) = 10\log_{10}\left(\frac{\text{mnk}\left\|\mathcal{H}\right\|_\infty^2}{\left\|\mathcal{X} - \mathcal{H}\right\|_F^2}\right) \tag{20}$$

$$\text{SSIM}(\mathcal{X}, \mathcal{H}) = \frac{(2\mu_\mathcal{X}\mu_\mathcal{H} + c_1)(2\sigma_{\mathcal{X}\mathcal{H}} + c_2)}{(\mu_\mathcal{X}^2 + \mu_\mathcal{H}^2 + c_1)(\sigma_\mathcal{X}^2 + \sigma_\mathcal{H}^2 + c_2)} \tag{21}$$

TABLE I
Baseline Approaches

Approaches	Description
HaLRTC[19]	A high accuracy low-rank tensor completion method, which is based on Tucker rank and implemented by the ADMM framework.
TNN[29]	A tensor completion method using tensor nuclear norm, this method is based on Tubal rank and implemented by the ADMM framework.
SiLRTC-TT[40]	A simple low-rank tensor completion method, this method is based on TT rank and implemented by the BCD framework.
TMac-TT[40]	A tensor completion method using parallel matrix factorization, this method is based on the TT rank and implemented by the BCD framework.
TMacTT-OKA[44]	A tensor completion method using parallel matrix factorization and overlapping ket augmentation, this method is based on the TT rank and implemented by the BCD framework.
FastHyIn[35]	An hyperspectral image inpainting method that utilizes the self-similarity of subspace representation coefficients and uses SVD for low-rank representation.
NLFCTN[36]	A remote sensing image inpainting method that uses fully-connected tensor network decomposition to explore the low-rank properties of the whole image and its non-local self-similarity groups.

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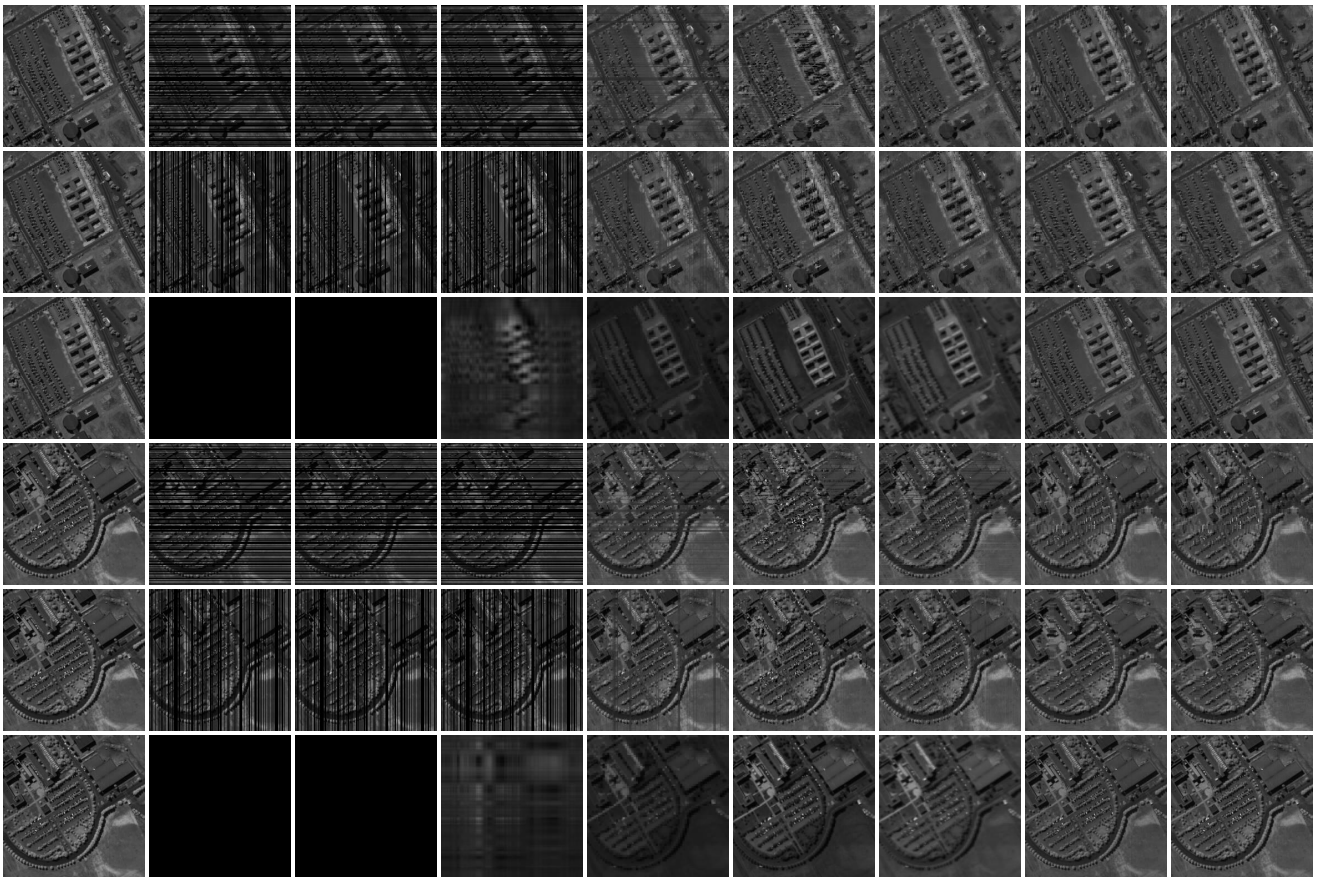

Fig. 4. 80th band of HSIs restored by various methods with 50% random stripe missing and fixed spectral missing on the Pavia University dataset.

where $\mu_{\mathcal{X}}$ and $\mu_{\mathcal{H}}$ are the average values of $\mathcal{X}$ and $\mathcal{H}$, respectively, $\sigma_{\mathcal{X}}^2$ and $\sigma_{\mathcal{H}}^2$ are the variances of $\mathcal{X}$ and $\mathcal{H}$, respectively, $\sigma_{\mathcal{X}\mathcal{H}}$ is the covariance of both, and c_1 and c_2 are constants used to maintain stability.

B. Experimental Results

1) Pavia University Dataset: This section compares the effectiveness of various tensor completion algorithms on the Pavia University dataset. The original size of the remote sensing image is 610 × 610 × 103. For the convenience of experiments and analysis, two non-overlapping images with a size of 256 × 256 × 81 were selected and named PU1 and PU2, respectively. The image is reshaped to $12\times12\times12\times12\times256$, $12\times12\times12\times12\times256$, and $16\times16\times16\times16\times81$ along each mode. For Cases 1 and 2, the missing rates were set to 0.5, 0.7, and 0.9, respectively; For Case 3, set 10, 20, 30, 40, 50, 60, 70, and 80 as the missing bands.

Qualitative comparison: Fig. 4 shows the 80th band of HSIs PU1 and PU2 restored by various methods in various scenarios, with a missing rate of 50% for Cases 1 and 2. In the scenario of stripes missing, HaLRTC and TNN lose their effectiveness completely because these two methods do not involve TA and complete the missing data in the original third-order space. When one or more slice matrices of the tensor are completely missing, the correlation of the data cannot be obtained. SiLRTC-TT and TMac-TT augment the third-order tensor to a higher-order one along the mode-3 direction and then use the TT unfolding matrix of the higher-order tensor to better capture the spatial correlation of the image. Compared with the first two algorithms, the effectiveness was improved. On the basis of TMac-TT, TMacTT-OKA uses overlapping KA to augment the tensor, eliminating block artifacts and further improving the effectiveness. In the scenario of spectral missing, TNN has a certain inpainting effect compared with HaLRTC, because the algorithm performs Fourier transform along mode-3 direction, which can capture the correlation between bands to a certain extent. The SiLRTC-FTT and TMac-FTT proposed in this article exhibit the best clarity in various situations.

Quantitative comparison: Table II reports the numerical results obtained through various methods on the Pavia University dataset, including PSNR, SSIM, and mean time. The best experimental results have been highlighted in bold black. Corresponding to the results in Fig. 4, the HaLRTC and TNN results are consistent, both in a completely unrestored state. SiLRTC-TT, TMac-TT, and TMacTT-OKA capture spatial correlation through mode-3 augmentation of tensors, and the numerical results show a certain improvement. It can be seen that the proposed SiLRTC-FTT algorithm achieves the best results in terms of quality evaluation indicators. In terms of algorithm efficiency, due to the two proposed methods augmenting the HSI on three modes, the cost is relatively high, especially for SiLRTC-FTT involving SVD. Fig. 5 shows the comparison of PSNR and SSIM for each missing band

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TABLE II

QUANTITATIVE COMPARISON OF ALL COMPARED METHODS FOR CASES 1 AND 2 OF PAVIA UNIVERSITY DATASETS

Dataset	Case	MR	Indexes	HaLRTC	TNN	SiLRTC -TT	TMac -TT	TMacTT -OKA	SiLRTC -FTT	TMac -FTT
PU1	Case1	0.5	PSNR	17.6739	17.6739	27.6269	24.6740	28.9902	**30.0266**	29.9263
			SSIM	0.2757	0.2757	0.8229	0.6853	0.8555	**0.8815**	0.8745
		0.7	PSNR	16.4385	16.4385	24.1722	22.2125	25.5894	**26.7814**	26.7219
			SSIM	0.1821	0.1821	0.7001	0.5384	0.7348	**0.7731**	0.7657
		0.9	PSNR	14.9593	14.9593	18.7708	19.4920	20.5527	**21.7933**	21.7370
			SSIM	0.0417	0.0417	0.3899	0.2328	0.3713	**0.4760**	0.4681
	Case2	0.5	PSNR	18.0199	18.0199	28.9360	25.9957	29.6031	**30.9229**	30.8148
			SSIM	0.3079	0.3079	0.8608	0.7769	0.8551	**0.8824**	0.8763
		0.7	PSNR	16.3891	16.3891	24.2401	22.1896	25.5930	**26.4351**	26.3834
			SSIM	0.1725	0.1725	0.6901	0.5291	0.7091	**0.7224**	0.7147
		0.9	PSNR	15.1392	15.1392	19.7593	19.4901	20.8831	**22.5636**	22.5203
			SSIM	0.0607	0.0607	0.4513	0.2498	0.3973	**0.5029**	0.4967
	Mean Time (S)			5	643	219	36	134	327	57
PU2	Case1	0.5	PSNR	18.3712	18.3712	27.6892	24.3395	27.1637	**29.2819**	29.1968
			SSIM	0.3092	0.3092	0.8302	0.6935	0.7811	**0.8771**	0.8708
		0.7	PSNR	16.5282	16.5282	23.4400	21.2204	23.6611	**25.6225**	25.5338
			SSIM	0.1728	0.1728	0.6505	0.3929	0.5978	**0.7294**	0.7215
		0.9	PSNR	15.4255	15.4255	20.4019	19.8315	20.5259	**22.2161**	22.1595
			SSIM	0.0504	0.0504	0.4768	0.2212	0.2901	**0.4896**	0.4841
	Case2	0.5	PSNR	18.5506	18.5506	28.6616	26.6931	28.9892	**29.9457**	29.8951
			SSIM	0.3345	0.3345	0.8631	0.8089	0.8514	**0.8923**	0.8879
		0.7	PSNR	16.6953	16.6953	24.2609	22.3325	24.8841	**26.2185**	26.1654
			SSIM	0.1760	0.1760	0.6787	0.5001	0.6587	**0.7437**	0.7381
		0.9	PSNR	15.4449	15.4449	20.0096	20.4195	21.1047	**22.4851**	22.4316
			SSIM	0.0541	0.0541	0.4571	0.2809	0.3651	**0.4980**	0.4915
	Mean Time (S)			5	624	218	33	132	324	56

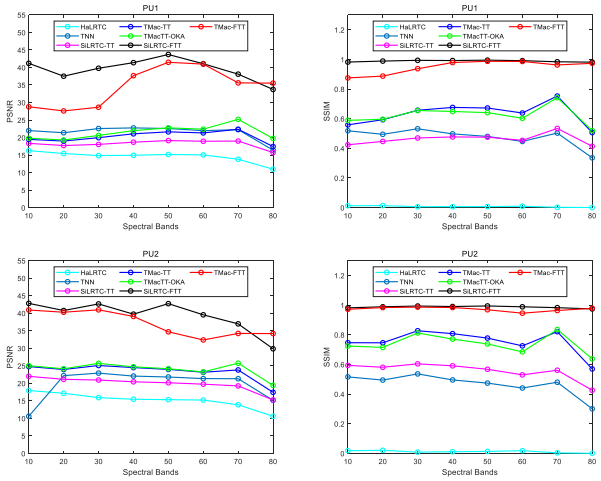

Fig. 5. Quantitative comparison of missing bands on the Pavia University dataset.

in Case 3. Compared with TMac-TT, the effectiveness of TMacTT-OKA is less improved in the scenario of spectral missing. This is because TMacTT-OKA still only augments the image in the spatial domain, resulting in insufficient mining of the correlation in the spectral domain. It can be seen that the two algorithms proposed in this article, SiLRTC-FTT and TMac-FTT, have much better performance than other algorithms, and their advantages are more obvious compared to stripe missing. This is because through mode-1 and -2 augmentation, the correlation between different bands of HSIs can be better captured.

2) Washington DC Mall Dataset: This section compares the effectiveness of various tensor completion algorithms on the Washington DC Mall dataset. The original size of the remote sensing image is 1208 × 307 × 210. For the convenience of experiments and analysis, two non-overlapping images with a size of 256 × 256 × 144 were selected and named WDC1 and WDC2, respectively. The image is reshaped to 2 × 2 × 4 × 4 × 4 × 4 × 6 × 6 × 256, 2 × 2 × 4 × 4 × 4 × 4 × 6 × 6 × 256, and 4 × 4 × 4 × 4 × 4 × 4 × 4 × 4 × 144 along each mode. For Cases 1 and 2, the missing rates were set to 0.5, 0.7, and 0.9, respectively; For Case 3, set 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, and 140 as the missing bands.

TABLE III

QUANTITATIVE COMPARISON OF ALL COMPARED METHODS FOR CASES 1 AND 2 OF WASHINGTON DC MALL DATASETS

Dataset	Case	MR	Indexes	HaLRTC	TNN	SiLRTC -TT	TMac -TT	TMacTT -OKA	SiLRTC -FTT	TMac -FTT
WDC1	Case1	0.5	PSNR	14.8992	14.8992	23.5673	20.6745	24.3967	**27.1114**	27.0040
			SSIM	0.2350	0.2350	0.7312	0.5953	0.7506	**0.8472**	0.8391
		0.7	PSNR	13.5699	13.5699	20.6328	18.7501	21.0803	**23.2000**	23.1161
			SSIM	0.1493	0.1493	0.5869	0.4380	0.5728	**0.7004**	0.6977
		0.9	PSNR	12.6212	12.6212	18.1119	17.5370	18.8448	**20.6313**	20.5006
			SSIM	0.0545	0.0545	0.3929	0.2651	0.3932	**0.5282**	0.5233
	Case2	0.5	PSNR	15.0528	15.0528	23.0101	20.1536	23.6585	**26.2076**	26.1346
			SSIM	0.2630	0.2630	0.7277	0.5806	0.7425	**0.8328**	0.8243
		0.7	PSNR	14.1913	14.1913	21.5478	18.8338	21.4331	**24.8909**	24.8517
			SSIM	0.1790	0.1790	0.6144	0.4434	0.5872	**0.7498**	0.7461
		0.9	PSNR	12.4965	12.4965	17.8436	16.9677	17.7060	**19.2897**	19.1484
			SSIM	0.0353	0.0353	0.3477	0.2373	0.3071	**0.4320**	0.4267
	Mean Time (S)			11	1204	836	69	248	1097	187
WDC2	Case1	0.5	PSNR	15.1990	15.1990	24.3061	21.7638	25.8063	**27.6163**	27.4803
			SSIM	0.2668	0.2668	0.7709	0.6239	0.7923	**0.8557**	0.8460
		0.7	PSNR	14.0787	14.0787	22.0362	19.7603	22.6597	**24.6910**	24.5965
			SSIM	0.1813	0.1813	0.6477	0.4882	0.6586	**0.7350**	0.7303
		0.9	PSNR	12.8467	12.8467	18.4305	17.8328	19.2878	**20.5940**	20.5011
			SSIM	0.0651	0.0651	0.4251	0.2729	0.4073	**0.5209**	0.5139
	Case2	0.5	PSNR	15.1459	15.1459	23.8483	21.3504	25.8545	**26.9746**	26.8699
			SSIM	0.2630	0.2630	0.7618	0.6252	0.7968	**0.8424**	0.8325
		0.7	PSNR	13.8056	13.8056	20.6166	18.9740	22.1359	**23.9308**	23.8589
			SSIM	0.1580	0.1580	0.5989	0.4337	0.6335	**0.7037**	0.7003
		0.9	PSNR	13.0384	13.0384	19.3454	17.5901	19.5961	**21.5152**	21.3955
			SSIM	0.0776	0.0776	0.4631	0.2497	0.4307	**0.5529**	0.5467
	Mean Time (S)			13	1223	825	67	245	1072	186

Qualitative comparison: Fig. 6 shows the 80th band of HSIs WDC1 and WDC2 restored by various methods in various scenarios, with a missing rate of 50% for Cases 1 and 2. It can be seen that in the scene of stripes missing, HaLRTC and TNN are completely unable to complete the missing entries, and SiLRTC-TT and TMac-TT are missing some local details. In the scenario of spectral missing, HaLRTC still completely loses its effectiveness, TNN has severe distortion, while SiLRTC-TT and TMac-TT are relatively blurry and have significantly darker colors. In various scenarios, the two proposed algorithms, SiLRTC-FTT and TMac-FTT, both exhibit outstanding visual effects.

Quantitative comparison: Table III and Fig. 7, respectively, show the numerical results of various algorithms in the scenario of stripe missing and spectral missing. Compared to other algorithms, whether PSNR or SSIM, the two proposed algorithms based on full-mode-augmentation TT rank minimization have significant advantages. Due to PSNR being based on errors between pixels, while SSIM evaluates the quality of the image from three aspects: brightness, contrast, and structure, there may be situations where the updates of the two metrics are not synchronized.

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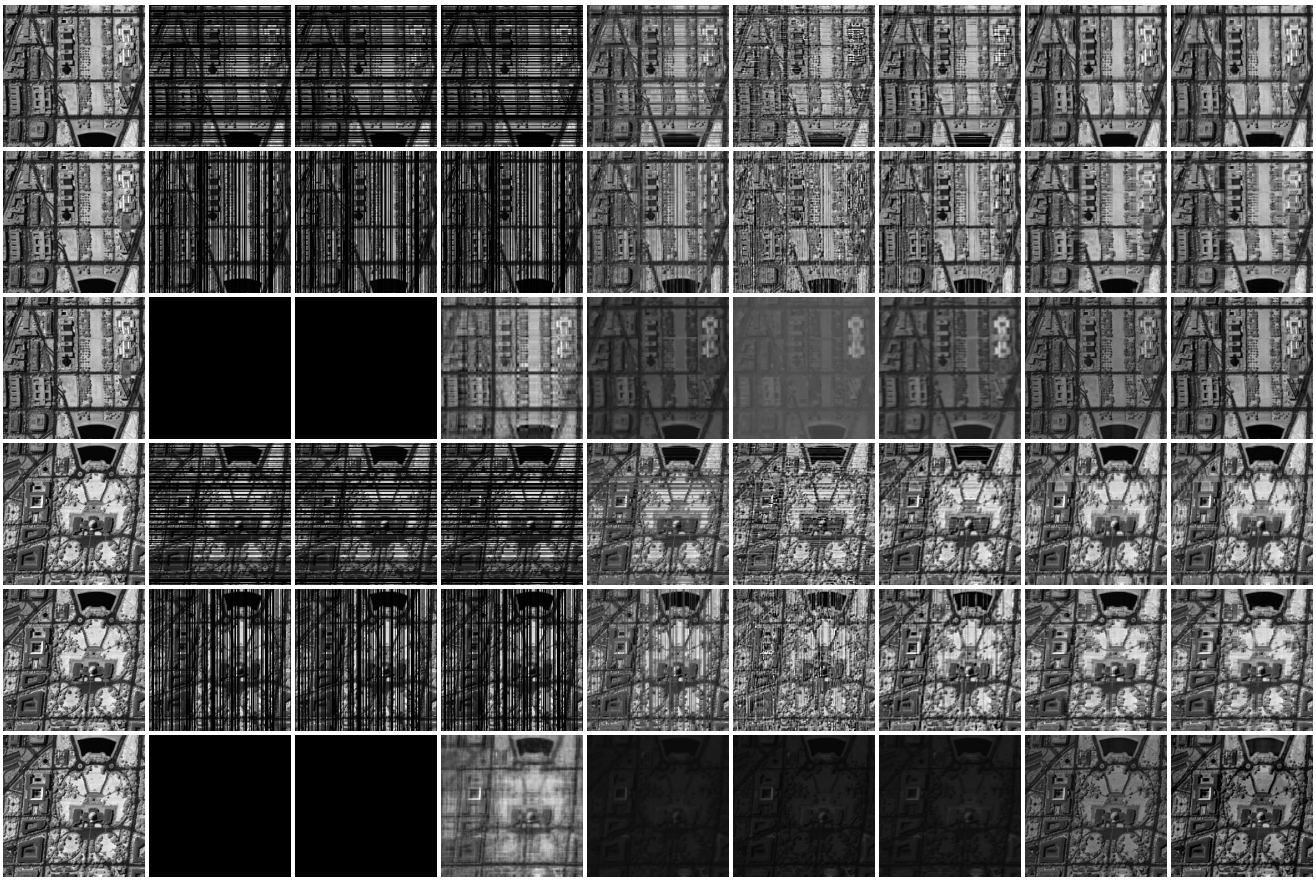

Fig. 6. 80th band of HSIs restored by various methods with 50% random stripe missing and fixed spectral missing on the Washington DC Mall dataset.

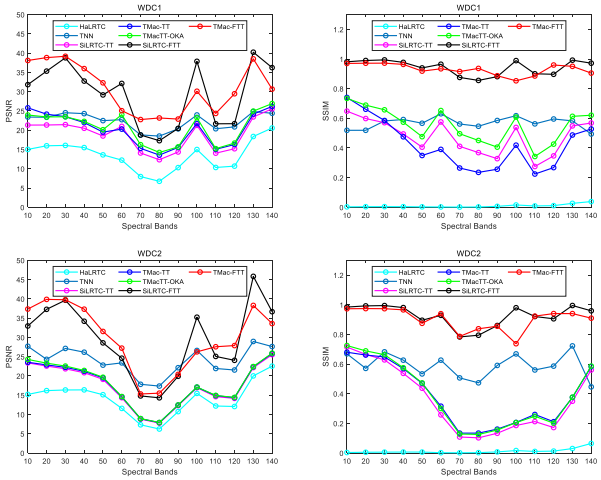

Fig. 7. Quantitative comparison of bands missing on the Washington DC mall dataset.

TABLE IV

QUANTITATIVE COMPARISON OF THE PROPOSED METHOD, HSI INPAINTING METHODS, AND OPTIMIZED HSI INPAINTING METHODS USING THE PROPOSED FRAMEWORK

Case	Indexes	FastHyIn	NLFCTN	TMac -FTT	FTT -FastHyIn	FTT -NLFCTN
Stripes Missing (50% Bands)	PSNR	42.7809	43.2645	47.6569	48.2643	48.5547
	SSIM	0.9854	0.9901	0.9951	0.9957	0.9962
Stripes Missing (All Bands)	PSNR	18.0150	18.0332	30.8153	30.9981	30.9993
	SSIM	0.3033	0.3076	0.8756	0.8829	0.8831
Spectral Missing	PSNR	14.2565	14.2625	31.7090	31.7142	31.7349
	SSIM	0.0073	0.0071	0.9471	0.9502	0.9514
Mean Time (S)		28	1935	53	67	1954

C. Discussion

1) Comparison With HSI Inpainting Methods and Their Improvement: We compared the proposed method with two low-rank-based HSI inpainting methods on the Pavia University dataset. The missing scenes include stripes missing of 50% bands, stripes missing of all bands, and spectral missing. According to Fig. 8 and Table IV, FastHyIn and NLFCTN are completely lost their effectiveness in scenarios of all bands stripes missing and spectral missing. This is because these two methods can only capture the correlation of data along a single mode. When the data is completely missing along a certain mode, the image cannot be restored. As a comparison, the proposed tensor completion method TMac-FTT using full-mode-augmentation has significant advantages.

In addition, we have improved these two HSI inpainting methods by optimizing their rank function using the proposed framework FTT($\mathcal{X}$), where FTT represents the full-mode-augmentation TTD and $\mathcal{X}$ represents the tensor of HSI. The experimental results show that the two optimized inpainting algorithms FTT-FastHyIn and FTT-NLFCTN have stronger ability for capturing the global low-rank structure of HSI, and have shown outstanding performance in all three scenarios.

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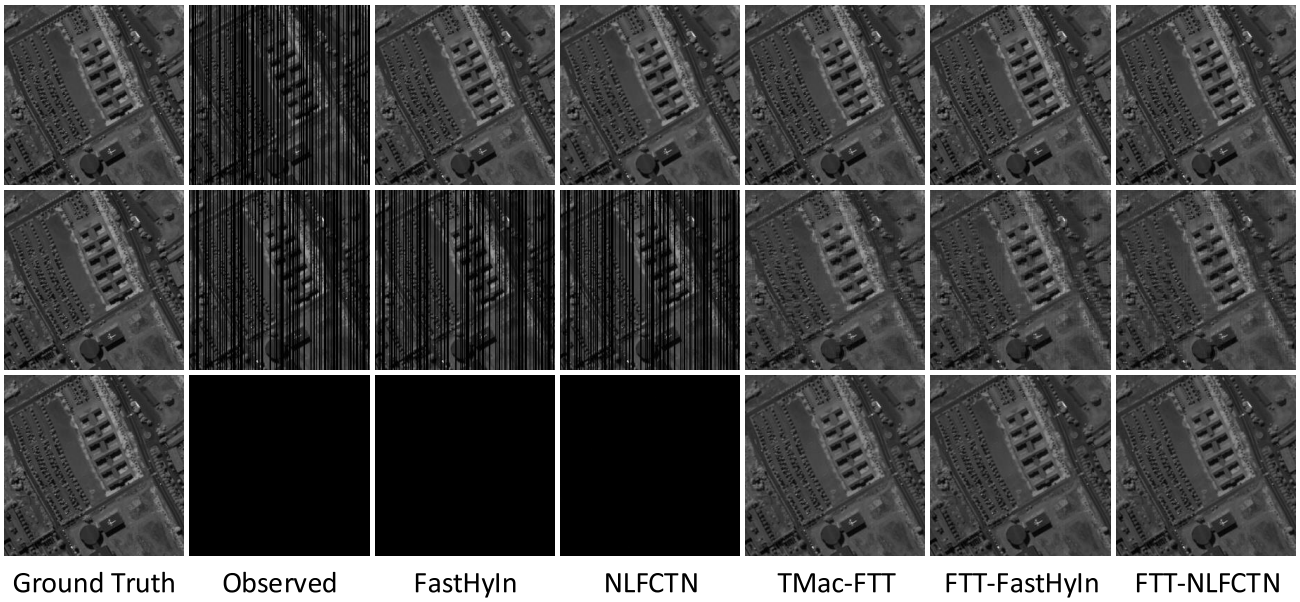

Fig. 8. Comparison with HSI inpainting methods and their improvement on the Pavia University dataset.

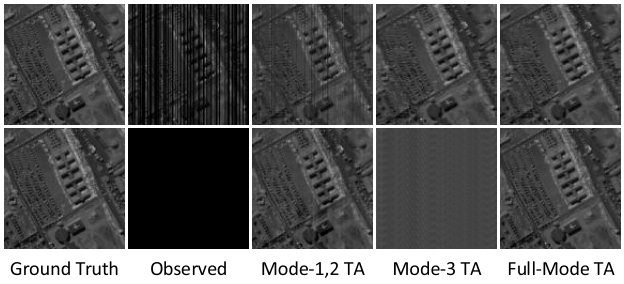

Fig. 9. Ablation experiment results on the Pavia University dataset.

TABLE V
QUANTITATIVE COMPARISON OF MODE-1,2 TA, MODE-3 TA, AND FULL-MODE TA IN TWO SCENARIOS

Case	Indexes	Mode-1,2 TA	Mode-3 TA	Full-Mode TA
Spatial Missing	PSNR	25.6237	29.6428	29.7347
	SSIM	0.6780	0.8379	0.8438
Spectral Missing	PSNR	28.3516	20.1657	31.7214
	SSIM	0.9079	0.4483	0.9478
Mean Time (S)		43	31	54

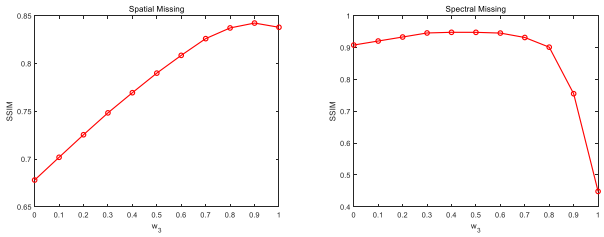

Fig. 10. Impact of parameter w_3 on the results in two scenarios.

2) Ablation Analysis: To analyze the impact of each mode TA on HSI inpainting, we conducted ablation experiments on the Pavia University dataset. The missing scenes include spatial missing (stripes missing) and spectral missing. Set the weights of each mode to $w_1 = \sum_{j=1}^{N-1} \alpha_{1j}$, $w_2 = \sum_{j=1}^{N-1} \alpha_{2j}$, and $w_3 = \sum_{j=1}^{N-1} \alpha_{3j}$. Considering that the horizontal space and vertical space of HSI have the same characteristics along the third mode. We assigned the same weight to mode-1 and -2, i.e., $w_1 = w_2 = (1 - w_3)/2$.

The results of the ablation experiment are shown in Fig. 9 and Table V. From this, it can be seen that spectral domain augmentation, i.e., mode-1, 2 TA ($w_1 = w_2 = 0.5, w_3 = 0$), is effective for spectral missing scenes, while spatial domain augmentation, i.e., mode-3 TA ($w_1 = w_2 = 0, w_3 = 1$), is effective for spatial missing scenes. In contrast, full-mode-augmentation achieved the best experimental results in both scenarios. Fig. 10 shows the impact of w_3 on the results in two scenarios, and we select appropriate weight parameters for full-mode-augmentation based on the results.

3) Order Analysis: The full-mode-augmentation of the tensor actually augments the order along each mode, and the determination of order is a hard problem. To study the effect of order on tensor completion, Fig. 11 shows the changes in PSNR and SSIM of SiLRTC-FTT and TMac-FTT with orders ranging from 3 to 15 in the Case 3 on the Washington DC Mall dataset. From the figure, it can be seen that when the original third order is augmented to fourth order, the effectiveness is significantly improved, while from the selected three bands, there is no obvious change characteristic in the 4th to 15th order. This indicates that the augment operation has a significant impact on the results, but there is no need to discuss which order to augment to.

4) Convergence Analysis: Although optimization algorithms based on BCD can effectively improve computational efficiency by processing large-scale problems in blocks, their convergence is still an unresolved issue. Therefore, we tested and verified the convergence of the proposed SiLRTC-FTT and TMac-FTT through numerical experiments. Fig. 12 displayed the relative change [see (19)] curves of the proposed

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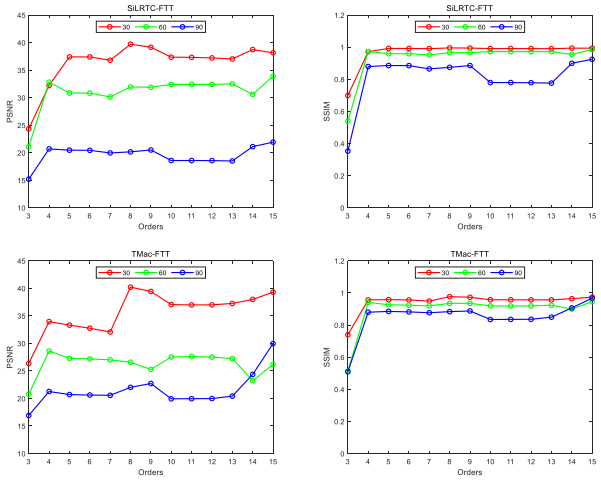

Fig. 11. Influence of order on algorithms SiLRTC-FTT and TMac-FTT.

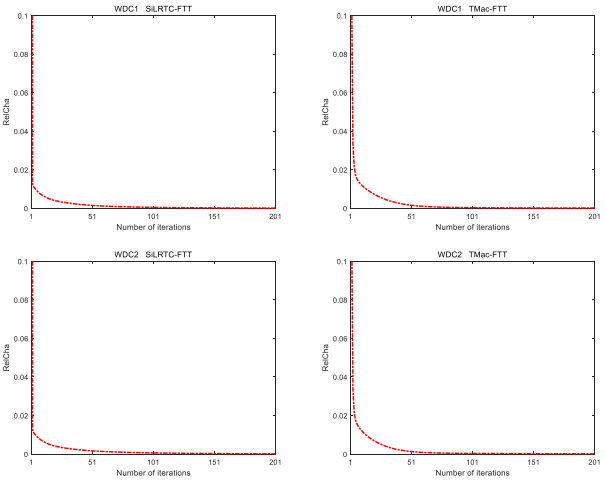

Fig. 12. Relative change curves of the BCD-based algorithms on the Washington DC mall dataset with the sampling rate 10%.

algorithms with respect to iterations on the Washington DC Mall dataset. We can evidently observe the numerical convergence of the algorithms.

V. Conclusion

In this article, we propose a tensor FTTD for HSI inpainting, and provide a tensor completion framework that achieves this decomposition by minimizing the rank of the unfolding matrix. Two widely used tensor completion algorithms, nuclear norm minimization and parallel matrix decomposition, were optimized using this framework. Numerous experiments have shown that the proposed method outperforms all compared methods.

In the future, we will attempt to combine the proposed tensor completion algorithms with various image priorities such as sparse, smooth, and self-similarity to further improve the effectiveness of image inpainting.

References

[1] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[2] G. Saiko, P. Lombardi, Y. Au, D. Queen, D. Armstrong, and K. Harding, "Hyperspectral imaging in wound care: A systematic review," *Int. Wound J.*, vol. 17, no. 6, pp. 1840–1856, Dec. 2020.

[3] S. Fang, K. Li, and Z. Li, "S^{2}ENet: Spatial–spectral cross-modal enhancement network for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[4] X. Sun et al., "Target detection through tree-structured encoding for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4233–4249, May 2021.

[5] C.-I. Chang, H. Cao, and M. Song, "Orthogonal subspace projection target detector for hyperspectral anomaly detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4915–4932, 2021.

[6] M. Imani and H. Ghassemian, "An overview on spectral and spatial information fusion for hyperspectral image classification: Current trends and challenges," *Inf. Fusion*, vol. 59, pp. 59–83, Jul. 2020.

[7] R. Hang, Z. Li, Q. Liu, P. Ghamisi, and S. S. Bhattacharyya, "Hyperspectral image classification with attention-aided CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2281–2293, Mar. 2021.

[8] Z. Han, D. Hong, L. Gao, J. Yao, B. Zhang, and J. Chanussot, "Multimodal hyperspectral unmixing: Insights from attention networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524913.

[9] W. He, H. Zhang, and L. Zhang, "Total variation regularized reweighted sparse nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3909–3921, Jul. 2017.

[10] Y.-B. Zheng, T.-Z. Huang, X.-L. Zhao, Q. Zhao, and T.-X. Jiang, "Fully-connected tensor network decomposition and its application to higher-order tensor completion," in *Proc. AAAI*, vol. 35, no. 12, 2021, pp. 11071–11078.

[11] H. Zhang, X.-L. Zhao, T.-X. Jiang, M. K. Ng, and T.-Z. Huang, "Multi-scale feature tensor train rank minimization for multidimensional image recovery," *IEEE Trans. Cybern.*, vol. 52, no. 12, pp. 13395–13410, Dec. 2022.

[12] J. Xue, Y. Zhao, W. Liao, J. C. Chan, and S. G. Kong, "Enhanced sparsity prior model for low-rank tensor completion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 11, pp. 4567–4581, Nov. 2020.

[13] H. Xu, M. Qin, Y. Yan, M. Zhang, and J. Zheng, "Nonlocal B-spline representation of tensor decomposition for hyperspectral image inpainting," *Signal Process.*, vol. 206, May 2023, Art. no. 108888.

[14] Y. Chang, L. Yan, B. Chen, S. Zhong, and Y. Tian, "Hyperspectral image restoration: Where does the low-rank property exist," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6869–6884, Aug. 2021.

[15] J. Zheng, J. Jiang, H. Xu, Z. Liu, and F. Gao, "Manifold-based nonlocal second-order regularization for hyperspectral image inpainting," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 224–236, 2021.

[16] O. Sidorov and J. Y. Hardeberg, "Deep hyperspectral prior: Single-image denoising, inpainting, super-resolution," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshop (ICCVW)*, Oct. 2019, pp. 3844–3851.

[17] R. Wong, Z. Zhang, Y. Wang, F. Chen, and D. Zeng, "HSI-IPNet: Hyperspectral imagery inpainting by deep learning with adaptive spectral extraction," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4369–4380, 2020.

[18] Z. Lai, K. Wei, and Y. Fu, "Deep plug-and-play prior for hyperspectral image restoration," *Neurocomputing*, vol. 481, pp. 281–293, Apr. 2022.

[19] J. Liu, P. Musialski, P. Wonka, and J. Ye, "Tensor completion for estimating missing values in visual data," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 1, pp. 208–220, Jan. 2013.

[20] Z. Long, Y. Liu, L. Chen, and C. Zhu, "Low rank tensor completion for multiway visual data," *Signal Process.*, vol. 155, pp. 301–316, Feb. 2019.

[21] S. Gandy, B. Recht, and I. Yamada, "Tensor completion and low-n-rank tensor recovery via convex optimization," *Inverse Problems*, vol. 27, no. 2, Feb. 2011, Art. no. 025010.

[22] Q. Song, H. Ge, J. Caverlee, and X. Hu, "Tensor completion algorithms in big data analytics," *ACM Trans. Knowl. Discovery From Data*, vol. 13, no. 1, pp. 1–48, Feb. 2019.

[23] Y. Xu, R. Hao, W. Yin, and Z. Su, "Parallel matrix factorization for low-rank tensor completion," 2013, *arXiv:1312.1254*.

[24] J. D. Carroll and J.-J. Chang, "Analysis of individual differences in multidimensional scaling via an n-way generalization of 'Eckart–Young' decomposition," *Psychometrika*, vol. 35, no. 3, pp. 283–319, Sep. 1970.

[25] L. R. Tucker, "Some mathematical notes on three-mode factor analysis," *Psychometrika*, vol. 31, no. 3, pp. 279–311, Sep. 1966.

[26] M. E. Kilmer and C. D. Martin, "Factorization strategies for third-order tensors," *Linear Algebra Appl.*, vol. 435, no. 3, pp. 641–658, Aug. 2011.

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[27] Q. Zhao, L. Zhang, and A. Cichocki, "Bayesian CP factorization of incomplete tensors with automatic rank determination," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 9, pp. 1751–1763, Sep. 2015.
[28] J. Xue, Y. Zhao, S. Huang, W. Liao, J. C. Chan, and S. G. Kong, "Multilayer sparsity-based tensor decomposition for low-rank tensor completion," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 11, pp. 6916–6930, Nov. 2022.
[29] Z. Zhang and S. Aeron, "Exact tensor completion using t-SVD," *IEEE Trans. Signal Process.*, vol. 65, no. 6, pp. 1511–1526, Mar. 2017.
[30] P. Zhou, C. Lu, Z. Lin, and C. Zhang, "Tensor factorization for low-rank tensor completion," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1152–1163, Mar. 2018.
[31] X. Li, Y. Ye, and X. Xu, "Low-rank tensor completion with total variation for visual data inpainting," in *Proc. 31st AAAI Conf. Artif. Intell. (AAAI)*, 2017, pp. 2210–2216.
[32] Y.-B. Zheng, T.-Z. Huang, T.-Y. Ji, X.-L. Zhao, T.-X. Jiang, and T.-H. Ma, "Low-rank tensor completion via smooth matrix factorization," *Appl. Math. Model.*, vol. 70, pp. 677–695, Jun. 2019.
[33] X.-T. Li, X.-L. Zhao, T.-X. Jiang, Y.-B. Zheng, T.-Y. Ji, and T.-Z. Huang, "Low-rank tensor completion via combined non-local self-similarity and low-rank regularization," *Neurocomputing*, vol. 367, pp. 1–12, Nov. 2019.
[34] X.-L. Zhao, W.-H. Xu, T.-X. Jiang, Y. Wang, and M. K. Ng, "Deep plug-and-play prior for low-rank tensor completion," *Neurocomputing*, vol. 400, pp. 137–149, Aug. 2020.
[35] L. Zhuang and J. M. Bioucas-Dias, "Fast hyperspectral image denoising and inpainting based on low-rank and sparse representations," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 730–742, Mar. 2018.
[36] W.-J. Zheng, X.-L. Zhao, Y.-B. Zheng, and Z.-F. Pang, "Nonlocal patch-based fully connected tensor network decomposition for multispectral image inpainting," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[37] Y.-B. Zheng, T.-Z. Huang, X.-L. Zhao, Y. Chen, and W. He, "Double-factor-regularized low-rank tensor factorization for mixed noise removal in hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 12, pp. 8450–8464, Dec. 2020.
[38] C. Peng et al., "Hyperspectral image denoising using nonconvex local low-rank and sparse separation with spatial–spectral total variation regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3206783.
[39] I. V. Oseledets, "Tensor-train decomposition," *SIAM J. Sci. Comput.*, vol. 33, no. 5, pp. 2295–2317, Jan. 2011.
[40] J. A. Bengua, H. N. Phien, H. D. Tuan, and M. N. Do, "Efficient tensor completion for color image and video recovery: low-rank tensor train," *IEEE Trans. Image Process.*, vol. 26, no. 5, pp. 2466–2479, May 2017.
[41] M. Ding, T.-Z. Huang, T.-Y. Ji, X.-L. Zhao, and J.-H. Yang, "Low-rank tensor completion using matrix factorization based on tensor train rank and total variation," *J. Sci. Comput.*, vol. 81, no. 2, pp. 941–964, Nov. 2019.
[42] J.-H. Yang, X.-L. Zhao, T.-H. Ma, M. Ding, and T.-Z. Huang, "Tensor train rank minimization with hybrid smoothness regularization for visual data recovery," *Appl. Math. Model.*, vol. 81, pp. 711–726, May 2020.
[43] X.-L. Zhao, J.-H. Yang, T.-H. Ma, T.-X. Jiang, M. K. Ng, and T.-Z. Huang, "Tensor completion via complementary global, local, and nonlocal priors," *IEEE Trans. Image Process.*, vol. 31, pp. 984–999, 2022.
[44] Y. Zhang, Y. Wang, Z. Han, X. Chen, and Y. Tang, "Effective tensor completion via element-wise weighted low-rank tensor train with overlapping ket augmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 11, pp. 7286–7300, Nov. 2022.
[45] T. G. Kolda and B. W. Bader, "Tensor decompositions and applications," *SIAM Rev.*, vol. 51, no. 3, pp. 455–500, Aug. 2009.
[46] M. Ding, T.-Z. Huang, X.-L. Zhao, M. K. Ng, and T.-H. Ma, "Tensor train rank minimization with nonlocal self-similarity for tensor completion," *Inverse Problems Imag.*, vol. 15, no. 3, p. 475, 2021.
[47] J.-F. Cai, E. J. Candès, and Z. Shen, "A singular value thresholding algorithm for matrix completion," *SIAM J. Optim.*, vol. 20, no. 4, pp. 1956–1982, Jan. 2010.
[48] J. S. Bhatt and M. V. Joshi, *Regularization Hyperspectral Unmixing*. Bellingham, WA, USA: SPIE, 2016.

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